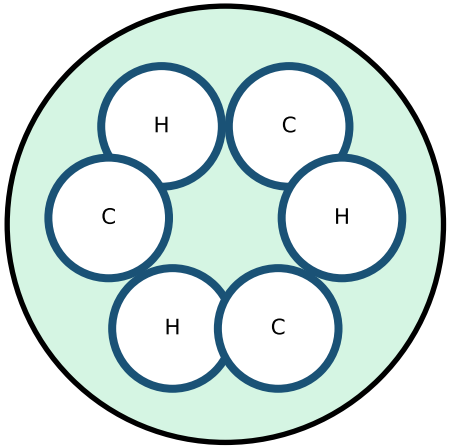
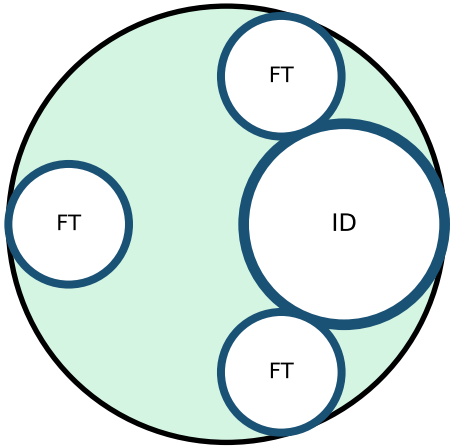
BOREAS v1.1 — 1. The geometry change, and why it was forced

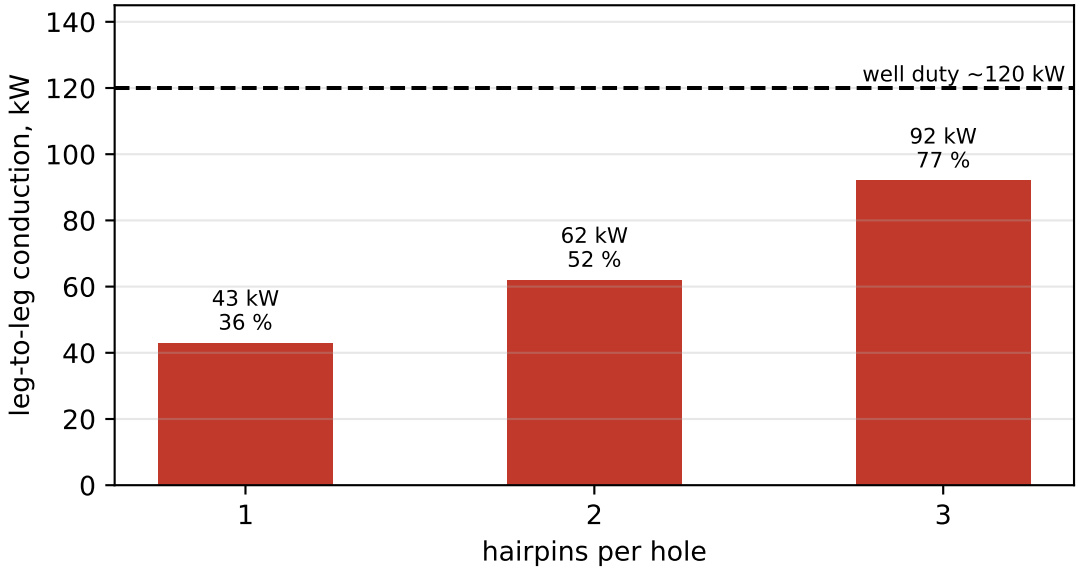
THUMS: 3 hairpins, 6 legs
H = hot leg, C = cold leg, SAME depth



BOREAS: 3 fin tubes + 1 insulated downcomer
all FT downward, ID returns



The term THUMS never had:
path grading short-circuits the cascade



Why the two changes are one change

A hairpin's return leg runs ~45 K BELOW the melting point, so the cascade must be graded along the TUBE PATH, not along depth.

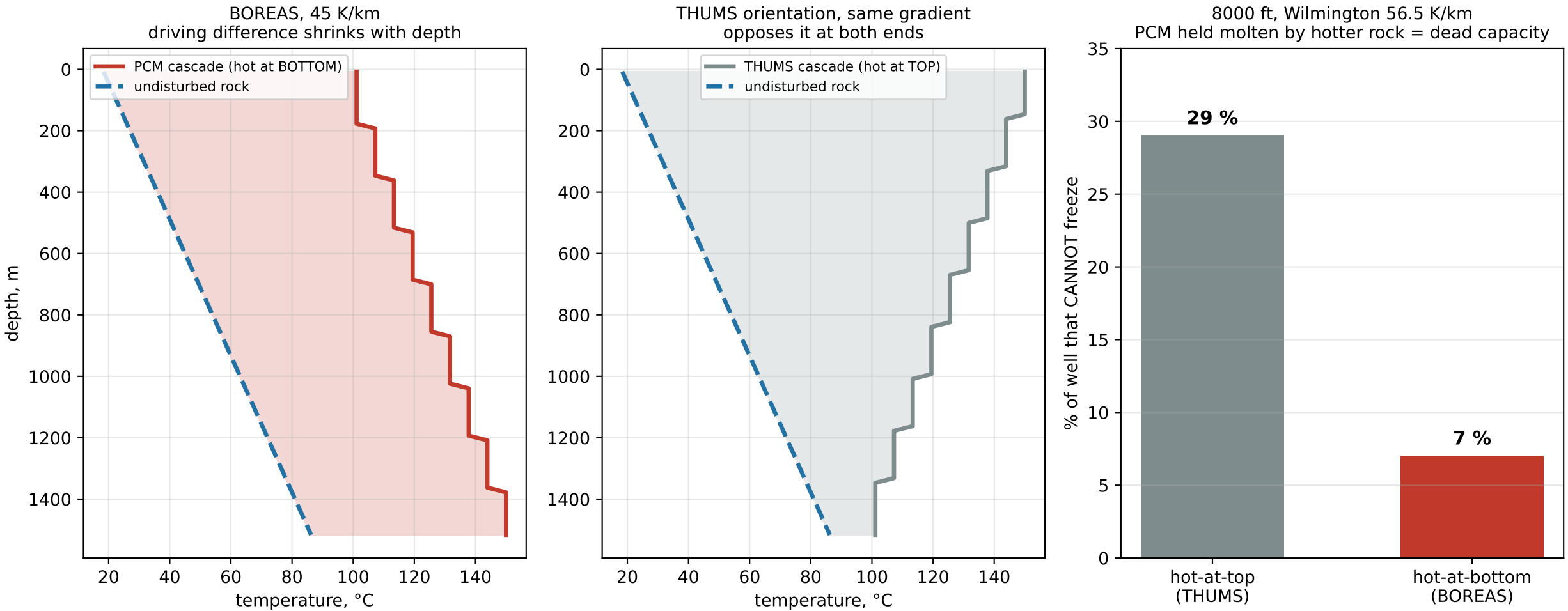
Two consequences:

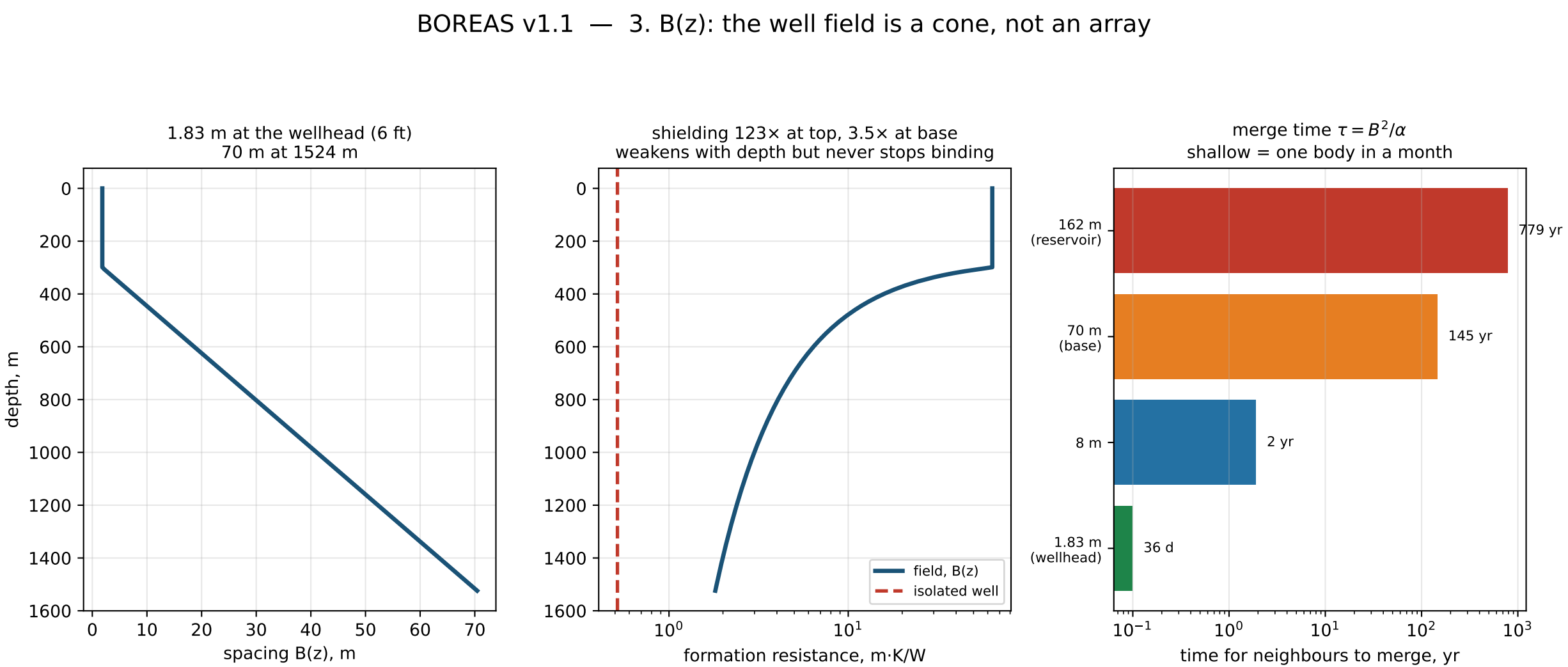
1. Two melting points coexist at one depth, in contiguous PCM with no wall between them. Conduction across that gap reaches 92 kW — 77 % of duty — and grows with hairpin count, the factor Study 2 ranked most beneficial.

2. z is a PATH LENGTH, so each depth is claimed by two tube segments. A depth-dependent rock temperature has two owners and no rule to split between them.

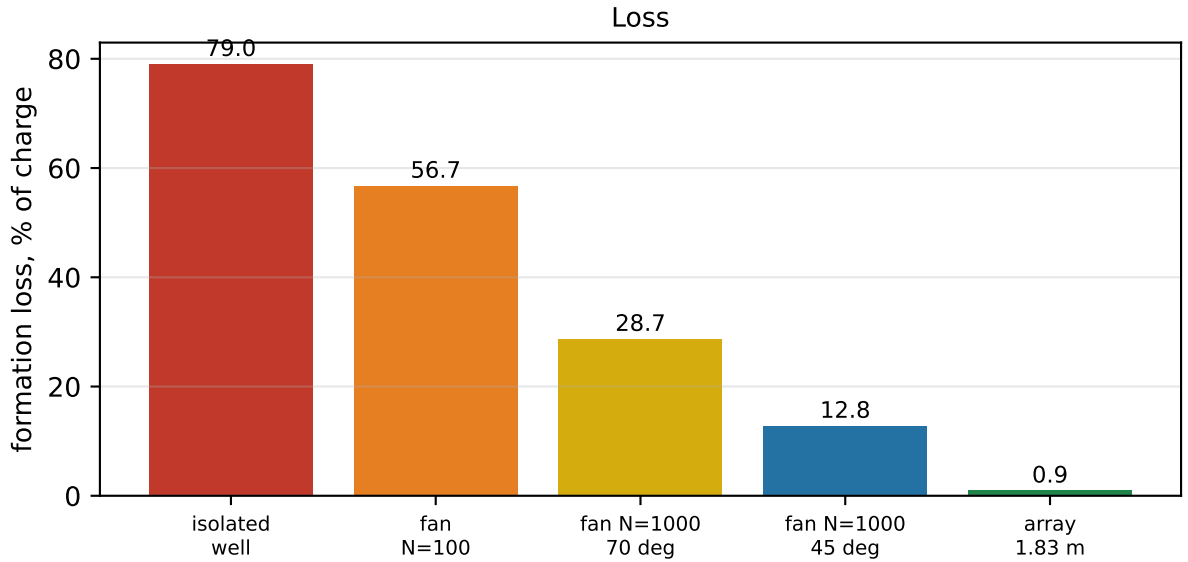
So the conducting wall could not be attached to the hairpin at all. The geometry had to change first.

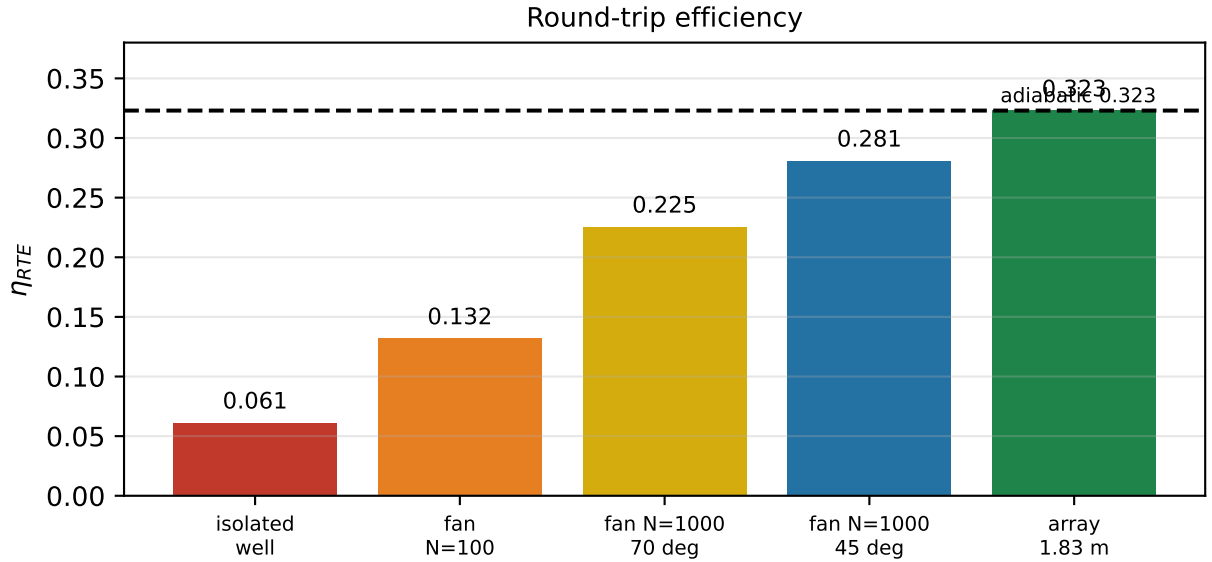
BOREAS v1.1 — 2. The cascade now aligns with the geothermal gradient

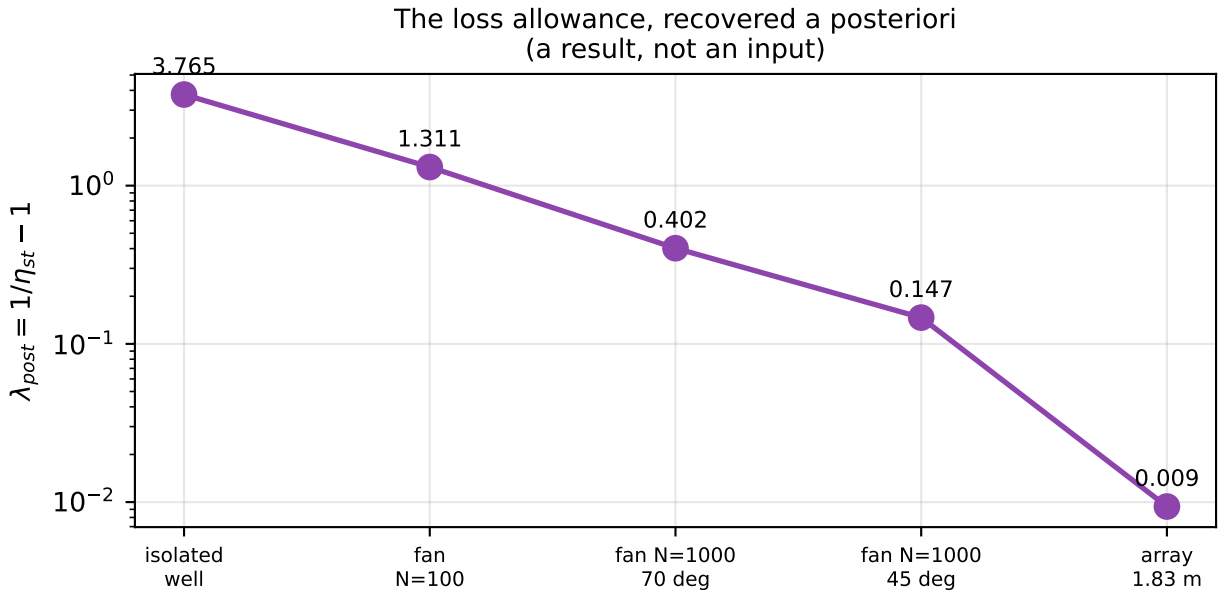



BOREAS v1.1 — 3. B(z): the well field is a cone, not an array
1.83 m at the wellhead (6 ft)
70 m at 1524 m
depth, m
spacing B(z), m
shielding 123× at top, 3.5× at base
weakens with depth but never stops binding
field, B(z)
isolated well
formation resistance, m·K/W
merge time $\tau = B^2/\alpha$
shallow = one body in a month
162 m (reservoir)
779 yr
70 m (base)
145 yr
8 m
2 yr
1.83 m (wellhead)
36 d
time for neighbours to merge, yr

BOREAS v1.1 — 4. Results: two orders of magnitude from one parameter







configuration	loss%	eta_st	etaRTE	eps	lam
isolated well	79.01	0.210	0.061	0.421	3.765
fan, N=100	56.73	0.433	0.132	0.582	1.311
fan, N=1000, 70 deg	28.69	0.713	0.225	0.773	0.402
fan, N=1000, 45 deg	12.78	0.872	0.281	0.868	0.147
array, 1.83 m	0.93	0.991	0.323	0.931	0.009

The isolated-well figure is NOT a pessimistic estimate of the loss. It is a different configuration, and one nobody would build.

The realistic case — 1000 wells fanning at 45° — loses 12.8 %, about 15 % relative on eta_RTE. Large enough that omitting it was a real defect; small enough that the concept survives it.

The fan ANGLE matters nearly as much as the well count, and it is set by where the reservoir lies: a SITE parameter, not a design one.

BOREAS v1.1 — 5. What the loss does to the store, by depth

